

DYNAMIC ENSEMBLE LEARNING FOR CREDIT CARD FRAUD DETECTION - KDD PROCESS REPORTs

BUSINESS UNDERSTANDING

DATA UNDERSTANDING: *Data Description Report*

- **Location:** The original dataset is available at <https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud>
- **Method used to acquire:** Download the dataset using the kagglehub library, which provides a simple way to interact with Kaggle resources.
- **Format:** csv file
- **Quantity:** Rows (284807) - Columns (31)
- **Identities of the fields:** The dataset contains transactions made by credit cards in September 2013 by European cardholders. This dataset presents transactions that occurred in two days. It contains only numerical input variables which are the result of a PCA transformation. Due to confidentiality issues, we cannot provide the original features and more background information. Features V1, V2, ... V28 are the principal components obtained with PCA, the only features which have not been transformed with PCA are 'Time' and 'Amount'. Feature 'Time' contains the seconds elapsed between each transaction and the first transaction in the dataset.
- **Surfaces features:** Amount, Time, PCs (V1 ... V28), Class
 - **Quantitative Variables**
 - Discrete: -
 - Continuous: Amount, Time, PCs (V1 ... V28)
 - **Categorical Variables**
 - Nominal: Class (0=Legit / 1=Fraud)
 - Ordinal: -

DATA UNDERSTANDING: *Data Quality Report*

- **Missing values Analysis**
 - **Column-Level Analysis:** Checked the presence of missing values in each of the columns to understand if some imputation was required. There are 0 columns with missing values.
 - **Row-Level Analysis:** Checked the presence of rows with more than 50% of missing values among the columns. There are 0 rows with more than 50% of missing values among the columns.
- **Class Balance:** The dataset presents a very high imbalance, as already known for frauds classification problems: Legit-0 (284315 – 99.827%) / Fraud (492 – 0.173%). In the next steps the data imbalance should be handled by using undersampling or oversampling techniques or by adopting models that can handle this very high imbalance within the dataset.
- **Duplicated Row:** The dataset contains 1081/284807 (0.380%) duplicated rows. In particular, it contains 773 duplicated patterns distributed as follows (611 different patterns repeated 2 times, 66 different patterns repeated 3 times, 81 different patterns repeated 4 times, 10 different patterns repeated 5 times, 1 pattern repeated 6 times, 2 different patterns repeated 9 times, 2 different patterns repeated 18 times). Additionally, we check duplicated both in terms of all features + Class and only looking to the features; in both cases we have 1081/284807 (0.380%).

If we leave duplicates in the dataset we may end up in:

 - Bias in the model - The classifier will “see” repeated evidence of the same row. This gives unfair influence to those examples during training.
 - Risk of overfitting - Especially with a small number of fraud cases, repeating exact Class 1 samples inflates the model's confidence.

- Invalidation of cross-validation - Duplicates might leak into both training and validation folds. This leads to optimistically biased performance metrics (especially precision/recall).

Based on these considerations, one solution could be of dropping duplicated rows, keeping each of them just one time.